PRD BILLING BABA

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Project Name: Billing Baba (Desktop B2B POS & Management System)
Platform: Desktop Application (Windows & macOS) optimized for offline-first resilience with cloud-sync + Mobile (Android & iOS).
Target Audience: Indian Retailers, Wholesalers, Service Providers, and Micro-Manufacturers.

1. Executive Summary & Vision

Billing Baba is engineered to be the operational brain for local shopkeepers and SMEs. As DevInDays focuses on capturing rapid adoption and reliable cash flow within the Indian market, this software bridges the gap between traditional, standalone billing software (like Vyapar) and modern, connected quick-commerce.
It provides an intuitive, Vyapar-inspired UI to ensure zero learning curve, but supercharges it with dynamic tier-based pricing, visual inventory mapping, deep CRM features, and a seamless, real-time sync to the Nazdikwala hyperlocal ecosystem. It is an offline-first desktop application that gracefully handles spotty internet while keeping online catalogs perfectly updated.

2. Global Dashboard & UI/UX (The Nerve Center)

The dashboard is built for extreme speed, minimizing clicks for the 50+ transactions a shopkeeper executes daily.

2.1 The "Omni-Search" & Quick Actions

- **1-Click Action Bar:** Fixed top-nav buttons to instantly trigger modals: *Create Invoice (Alt+N), Add Purchase (Alt+P), Add Item, Add Customer, Record Payment.*
- **Global Transaction Search:** A singular, lightning-fast search bar querying across Sales, Purchases, Payments, Customers, and Products via Name, Mobile Number, or Invoice Number.

2.2 Unified Business Overview

- **Financial Snapshot:** Real-time calculation of **Total Receivable** (Cash to collect) vs. **Total Payable** (Cash owed to suppliers). Clicking these metrics routes directly to the detailed Party Ledger.
- **Recent Orders Panel (Omni-Channel):** A unified feed combining:
  - 🟢 *Online Orders* (Incoming from Nazdikwala – requires Accept/Reject).
  - 🧾 *Offline Billing* (Over-the-counter walk-ins).
- **Sync Engine Indicator:** A persistent header icon showing cloud status: *"🟢 Synced with Nazdikwala (Last updated 2 sec ago)"* or *"🟠 Offline (Saving locally)"*.

2.3 Proactive Management Tools

- **Actionable To-Do List (Core USP):** A built-in task manager for daily operations (e.g., "Follow up with Sharma for ₹5,000," "Dispatch evening deliveries"). Includes priority tagging and completion checkboxes.

- **Smart Alerts Panel:** Proactive system warnings: *"₹50,000 payment overdue today,""Aashirvaad Atta stock below minimum level."*
- **Customizable Widgets (Drag & Drop):** Modular cards for Sales (Today/Month), Profit Summary, Top Selling Products, and Cash Flow, featuring mini sparkline graphs for instant visual trend analysis.

3. Core Accounting & Billing Workflows (The "Vyapar" Parity)
To compete effectively, Billing Baba must flawlessly execute standard Indian B2B accounting workflows.

3.1 The Sales Lifecycle (Outward Flow)

- **Estimates / Quotations:** Generate non-binding quotes. Stock is *not* deducted. Includes a 1-click "Convert to Invoice" button.
- **Delivery Challans:** Crucial for Indian B2B. Used when goods are dispatched but not yet billed (e.g., sending items for approval or in batches). Stock is deducted, but financial accounts are not hit. Includes a 1-click "Convert to Invoice" button.
- **Tax Invoices & Bill of Supply:** The core billing engine. Lightning-fast barcode scanner support. Auto-calculates CGST/SGST/IGST based on the customer's state code.
- **Sales Returns (Credit Notes):** Issued when a customer returns goods. Automatically adds stock back to inventory and adjusts the customer's outstanding balance.
- **POS Interface:** A dedicated, clutter-free "Point of Sale" mode designed entirely for keyboard-only operation (Tab, Enter, F-keys) for high-speed retail checkout.

3.2 The Purchase Lifecycle (Inward Flow)

- **Purchase Orders (PO):** Sent to suppliers to request stock.
- **Purchase Bills:** Recording incoming inventory. Automatically updates the average moving purchase price and increases stock levels.
- **Purchase Returns (Debit Notes):** Issued when returning damaged goods to a supplier. Deducts stock and adjusts payable accounts.
- **Expense Tracking:** Categorized tracking for operational costs (Electricity, Rent, Staff Salaries, Tea/Snacks) to calculate accurate Net Profit.

4. Parties Module (Smart CRM & Credit Management)

Upgrading from a simple digital ledger to a highly contextual Customer Relationship Management tool.

4.1 Onboarding & Organization

- **Quick Creation:** 1-click import from phone contacts or Gmail.
- **Area Tagging:** Assign customers to micro-regions (e.g., "Station Road") to optimize batch deliveries and generate area-based sales heatmaps.
- **Multi-Select Party Type:** Categorize as Customer, Retailer, Wholesaler, Manufacturer, or Custom.
- **Custom Tagging:** Apply operational labels (*VIP, Risky, New*) for quick visual identification.

- **Party Linking:** Group multiple profiles (e.g., connecting a father and son) to track shared family credit limits.

4.2 Deep Relationship Tracking

- **Business Lifetime Value (LTV):** Displays "Total Business Value" generated since inception (e.g., "Connected since 2022").
- **Smart Unpaid Tracking:** Accurately calculates true recoverable amounts, automatically factoring in partial payments and applied discounts.
- **Visit Tracking System:** Logs every interaction. Tracks Date, Time, Sale Type, and Purpose (*Purchase, Inquiry, Payment Collection, Casual Visit*).
- **File Attachments:** Snap and attach photos of Visiting Cards, GST Certificates, ID proofs, or contracts directly to the profile.
- **Face Image Capture:** Store customer photos for visual identification, laying the groundwork for future facial-recognition checkout.

4.3 Automated Financial Safety Nets

- **Credit Limits:** Define a maximum credit ceiling per party. The system hard-blocks or issues a bypassable warning if a new invoice exceeds this limit.
- **1-Click Payment Reminders:** Automatically generate and send outstanding balance reminders via WhatsApp or SMS.
- **Loyalty Points System:** Configurable reward rules (e.g., ₹100 = 1 Point). Displays total points earned and available balance for redemption on the next invoice.

5. Items & Inventory Module (The Smart Catalog)

Built to eliminate customer bargaining, manage complex warehouse logic, and sync effortlessly to the cloud.

5.1 Basic Setup & Cataloging

- **Bulk Upload:** CSV/Excel import mapping for rapid onboarding of thousands of items.
- **Product/Service Toggle:** Differentiate between physical items (stock tracking enabled) and services (labor, repair - stock tracking disabled).
- **Media Management:** Upload, re-order, and delete multiple product images, prioritizing the primary image for the Nazdikwala consumer app.

5.2 The Advanced Pricing Engine (Core USP)

- **Market Price (Reference Point):** Displays the general market price on the invoice to build trust ("Look, the market price is ₹850, but I am giving it to you for ₹800"), preventing aggressive bargaining.
- **Tier-Based Pricing:** Auto-applies pricing based on the customer's assigned tier in the Parties module:
  - *Tier 1 (Normal Walk-ins):* ₹800
  - *Tier 2 (Regular Customers):* ₹770

    Tier 3 (Wholesalers / Relatives): ₹700

- **Strict "MRP-Only" Rule:** A checkbox for fixed-price items (e.g., Lay's chips, Cigarettes). Disables Tiers, Market Price, and Discounts, locking the billing price strictly to the printed MRP.
- **Price History Tracking:** Immutably logs every price change with a date stamp. Sellers can pull up a trend graph to justify price hikes to complaining customers (e.g., showing manufacturer cost increases over 2 years).

5.3 Advanced Stock & Location Management

- **Standard Tracking:** Opening stock, average purchase price, and automated low-stock alerts.
- **Visual Location Management:** Sellers can define exact physical locations (e.g., "Warehouse B, Rack 4, Shelf 2") and attach timestamped photos of the shelf. Drastically reduces onboarding time for new shop helpers.
- **Multi-Unit Conversions:** Handle complex B2B units natively (e.g., Purchase in "Boxes", Sell in "Pieces", where 1 Box = 12 Pieces).

6. Reporting & Compliance

Instant access to the data required for Indian tax compliance and business health monitoring.

- **GSTR Reports:** Automated generation of GSTR-1, GSTR-2, and GSTR-3B in standard formats for direct portal upload or sharing with a Chartered Accountant.
- **Profit & Loss Statement:** Real-time P&L factoring in sales margins and recorded expenses.
- **Stock Summary & Valuation:** Live tracking of current inventory value based on FIFO or Average Cost methods.
- **Day Book:** A complete chronological ledger of every transaction, modification, and payment received on a given day for cash-drawer reconciliation.

7. Advanced B2B Operations & Quoting Engine

7.1 Bulk Brand Swapping (The "Alternative Finder")

The Problem: Generating a 20-item quotation for a premium brand (e.g., Havells), only for the customer to request a cheaper alternative. Retyping the entire quotation with a new brand wastes immense time. **The Solution:** The "Switch Brand" Bulk Action.

- **How it works:** A user selects all items on the current quotation and clicks "Find Alternatives." They select the target brand (e.g., "Orient").
- **The Logic:** The system maps the generic item category (e.g., 1.5mm Wire, 2-Way Switch) and automatically swaps the SKU to the Orient equivalent, preserving the exact quantities and instantly recalculating the total price.

7.2 Historical Price Conflict Alerts (The "Consistency Checker")

The Problem: In a multi-staff environment, Staff X sells an item to Customer A for ₹75. Two months later, Staff Y bills the identical item to Customer A for ₹80. The customer notices, feels cheated, and assumes Staff Y is overcharging. **The Solution:** Customer-Specific Pricing Guardrails.

- **How it works:** When adding an item to the cart, the system checks the Customer_ID and the Item_ID

against the transaction history.

- **The Alert:** If the entered price differs from their *last paid price*, a strict popup appears:
  - *"Warning: Customer previously paid ₹75 for this item on Jan 12th. Current price is ₹80. [Proceed Anyway] / [Match Previous Price]"*
- **The Benefit:** Enforces price consistency across all staff members, completely neutralizing customer disputes over fluctuating rates.

7.3 The "Kaccha to Pakka" Converter (Normal to GST Bill)

The Problem: Many Indian merchants do initial deals on plain paper or non-tax estimates, but later need to convert them into formal GST invoices for corporate clients or accounting purposes. **The Solution:** 1-Click Formalization.

- **How it works:** A dedicated toggle or button on saved "Normal Invoices/Estimates" that instantly duplicates the record, applies the correct intra-state (CGST/SGST) or inter-state (IGST) tax brackets, generates a sequential formal Invoice Number, and adjusts the ledger without messing up the backend inventory count.

8. HR & Staff Management Module

Billing Baba acts as the complete shop management suite, entirely eliminating the need for separate attendance registers.

8.1 Time & Attendance Tracking

- **Clock-In/Out:** Staff can log in using a designated system.
- **Shift Management:** Tracks late arrivals, half-days, and overtime hours automatically.

9. AI Automation & Data Entry (Phase 2 & Beyond)

9.1 AI Document Parsing (Smart Autofill)

- **The Workflow:** A vendor receives a 4-page printed purchase bill from a distributor. Instead of manually typing 50 items into the system to update stock, they upload a photo or PDF of the bill.
- **The Tech:** Optical Character Recognition (OCR) combined with AI extracts the Item Names, Quantities, HSN Codes, and Purchase Prices, auto-filling the "Add Purchase" screen in seconds.

9.2 The "Karma Score" & Defaulter Database

- **The Feature:** A shared ecosystem trust metric. If a B2B party consistently delays payments or bounces cheques, their profile gets tagged.
- **The Karma Score:** A creditworthiness rating visible to the seller before they issue new goods on credit, significantly reducing bad debt risk.

9.3 AI Voice Calling for Payment Recovery

- **The Workflow:** Instead of the shop owner wasting hours calling people for pending payments, the system triggers an AI Voice Agent.

- **The Execution:** The AI bot calls defaulters in native languages (Hindi/English), politely reminding them of the specific invoice amount and due date, and automatically logs their response (e.g., "Customer promised to pay on Monday") directly into the To-Do list on the dashboard.

10. Omni-Channel API Gateway

Billing Baba will not operate in a silo. The database architecture is built to act as the central source of truth for third-party integrations.

- **E-Commerce Aggregation:** Open API webhooks that allow real-time order pulling and inventory syncing with Swiggy, Zomato, Amazon, and Flipkart.
- **The Logic:** If a bottle of ketchup is sold offline at the POS, the stock automatically drops by 1 on Swiggy and Nazdikwala instantly, preventing the nightmare scenario of accepting an online order for an item that just sold out in the physical store.

11. The "Hidden Ledger" (Unofficial Recovery System)

The Problem: In offline Indian retail, customers frequently "round off" their payments (e.g., handing over exactly ₹10,000 for a ₹10,150 bill). Sellers absorb this to avoid an awkward argument and maintain the relationship. However, standard billing software forces this ₹150 into an official "Pending Due" balance, which shows up on the customer's next invoice and angers them. **The Solution:** A hidden, seller-only balance created from short payments, recoverable manually via future price overrides.

11.1 The Capture Mechanism (Short Payment Handling)

When a seller is recording a payment at checkout and the Paid Amount < Invoice Total:

- **The Smart Trigger:** The system intercepts the transaction and displays a prompt: *"Remaining ₹150 detected. How do you want to handle it?"*
- **The Options:** 1. **Add to Payment Balance (Official):** Becomes standard debt, prints on the next invoice. 2. **Add to Unrecovered Amount (Unofficial):** Sweeps the ₹150 into the "Hidden Ledger."

11.2 Storage & Strict Visibility Rules

This data is highly sensitive to the customer-seller relationship.

- **Storage:** The ₹150 is added to a hidden unrecovered_amount field on the customer's CRM profile. This amount compounds over time (e.g., ₹150 + ₹200 = ₹350).
- **Visibility Constraints:** * ❌ **Invisible to Customer:** Will *never* print on an invoice, will *never* appear in the Nazdikwala consumer app, and does not show up in the customer-facing ledger statement.
  - ✅ **Visible to Seller:** Only appears as an internal metric on the seller's dashboard and the customer's backend profile.

11.3 The Recovery Workflow (Silent Recapture)

Sellers traditionally recover these losses by slightly inflating the price of an item the next time the customer visits. The system actively assists with this.

- **The Action:** The customer returns. The seller adds an item to the cart and manually overrides/increases the price slightly above the standard tier rate.
- **The Smart Prompt:** The system detects the manual markup and checks the Hidden Ledger. A discreet popup appears for the seller: *"This customer has ₹700 unrecovered amount. Recover now?"*
- **The Resolution Options:**
  - **[Recover Full]:** System silently deducts the full amount from the hidden ledger (unrecovered_amount = 0).
  - **[Recover Partial]:** Seller inputs the recovered amount (e.g., ₹200), and the system updates the remaining hidden balance to ₹500.
  - **[Ignore]:** The markup is treated as standard profit, and the hidden ledger remains untouched.

11.4 Core System Rules

- **No Automatic Recovery:** The system will never automatically add this charge to a bill. The seller must maintain 100% manual control to protect the relationship.
- **No Audit Trail:** The customer's invoice simply shows the items at the price the seller entered. There is no line item that says "Previous Due Added" or "Adjusted."